Supporting Information

MaDDOSY (Mass Determination Diffusion Ordered Spectroscopy) using a 80 MHz Bench Top NMR for the Rapid Determination of Polymer and Macromolecular Molecular Weight

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Materials and Instrumentation

Poly(isoprene), poly(styrene sulfonate), poly(methyl methacrylate) and poly(styrene) standards were purchased from Polymer Standards Service (PSS) GmbH. Poly(ethylene glycol), Poly(amidoamine) dendrimers, d-chloroform, d-methanol, d-tetrahydrofuran, d-acetone, deuterium oxide, methyl acrylate, 1,4-dioxane, Azobisisobutyronitrile (AIBN), ascorbic acid, tert-Butyl hydroperoxide, 1-Naphthalenemethanol, L-lactide, anhydrous toluene, tin octoate, ethyl 2-bromoisobutyrate (EBiB), $CuBr_2$, Poly(propylene glycol) 2000, Isophorone diisocyanate, dibutyltin dilaurate, cyclohexane, tetrahydrofuran, dimethylsulfoxide, dimethylformamide, poly(styrene-co-maleic anhydride) cumene terminated, lysozyme, calcitonin, poly(vinyl pyridine), styrene-ethylene-butadiene-styrene and isoprene were purchased from Merck. Additional samples of styrene-ethylene-butadiene-styrene were also purchased from both Kraton and Polymer Source.

The RAFT agent, 2-(((butylthio)carbonothioyl)thio)propanoic acid (PABTC), was synthesised in-house, according to previously reported literature.[1] Tris[2-(dimethylamino)ethyl]amine (Me_6Tren) was synthesised according to previously reported literature.[2]

All DOSY NMR spectra were acquired on a Magritek Spinsolve 80 Carbon Benchtop NMR spectrometer equipped with a z-axis gradient coil capable of generating a maximum field gradient of 210 mT/m. For DOSY Measurements, the PGSTE pulse sequence was used, with the number of gradient steps, acquisition time, repetition time, number of scans, maximum gradient, little delta and big delta parameters optimised for each sample to ensure complete attenuation of the signal at the maximum gradient, with resolvable polymer peaks in all other steps. Specific parameters can be found in the raw data files at: https://wrap.warwick.ac.uk/. DOSY NMR spectra were interpreted using Magritek Spinsolve version 2.1.3. and the GNAT, version 1.3.2, developed by the Manchester NMR Methodology Group.[3] 1D ^{1}H NMR spectra were either acquired on a Magritek Spinsolve 80 Carbon Benchtop NMR spectrometer or a Bruker Avance III HD 400 MHz NMR Spectrometer.

GPC samples were run on an Agilent Infinity II MDS instrument equipped with differential refractive index (DRI), viscometry (VS), dual angle light scatter (LS) and multiple wavelength UV detectors. The system was equipped with 2 x PLgel Mixed C columns (300 x 7.5 mm) and a PLgel 5 µm guard column. The eluent was either $CHCl_3$, or THF with 0.01% BHT additive. Samples were run at 1 ml/min at 30 °C. Poly(methyl methacrylate), and polystyrene standards (Agilent EasyVials) were used for calibration. Ethanol was added as a flow rate marker. Analyte samples were filtered through a GVHP membrane with 0.22 µm pore size before injection. Respectively, experimental molar mass ($M_{n, SEC}$) and dispersity (Đ) values of synthesized polymers were determined by either conventional or universal calibration using Agilent GPC/SEC software.

Concentration Studies

The polymers of interest were dissolved in d-chloroform at the concentrations outlined in Figure 1 of the main text. In each case, 0.5 mL of the resultant solution was transferred into a standard NMR tube for measurement.

Solvent Independence Studies

Polystyrene GPC standards with molecular weights 1200, 19880, 52500 and 280500 and PMMA GPC standards with moleculaer weights 3790, 90250 and 342900 were each dissolved to concentrations of 7 mg mL^{-1}, after which 0.5 mL of the resultant solutions was transferred to a standard NMR tube for analysis.

Calibration

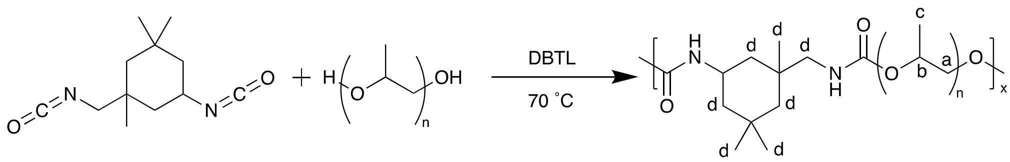
The calibrants used to build the calibration curve are detailed in table 1. The M_n used is that reported on the CoA from the respective standard. In each case, the calibrant was dissolved in the corresponding solvent, to a concentration of 7 mg mL^{-1}, after which 0.5 mL of the resultant solution was transferred into a standard NMR tube.

Table 1 - Polymers, Molecular Weights and Solvents used for Calibration.

Polymer	M_n / Da	Solvent
Poly(Amido Amine) Dendrimers	516.68, 1429.85, 3256.18, 6908.84, 14214.17	d-methanol
Polyisoprene	1070, 2190, 4610, 12000, 20400, 42500, 111000, 231000	d-chloroform
Poly(Methyl Methacrylate)	1200, 3790, 90250, 342900	d-chloroform
Polystyrene	1200, 19880, 52500, 215000, 280500	d-chloroform
Poly(Styrene Sulfonate)	1100, 1830, 4230, 10600, 29100	Deuterium oxide
Poly(Ethylene Glycol)	600, 1000, 2000, 8000	d-methanol

Polyurethane Synthesis

Poly(propylene glycol) 2000 (3.212 g, 1.61 mmol, 1 eq.) was added to Schlenk tube with a stirrer bar and placed under a nitrogen atmosphere. Isophorone diisocyanate (0.340 mL, 1.60 mmol, ~1 eq.) was added followed by dibutyltin dilaurate (5 drops, cat.). The reaction was stirred, heated to 70 °C, and left for 3 hours + 20 minutes. The product was obtained as a colourless semisolid. SEC (THF): M_n = 29,800 Da, M_w = 99,300 Da, $Đ_M$ = 3.33, MaDDOSY M_{wt} = 101,000. SEC Values obtained using PMMA calibration on RI detector.



Scheme 1 - Synthesis of polyurethane

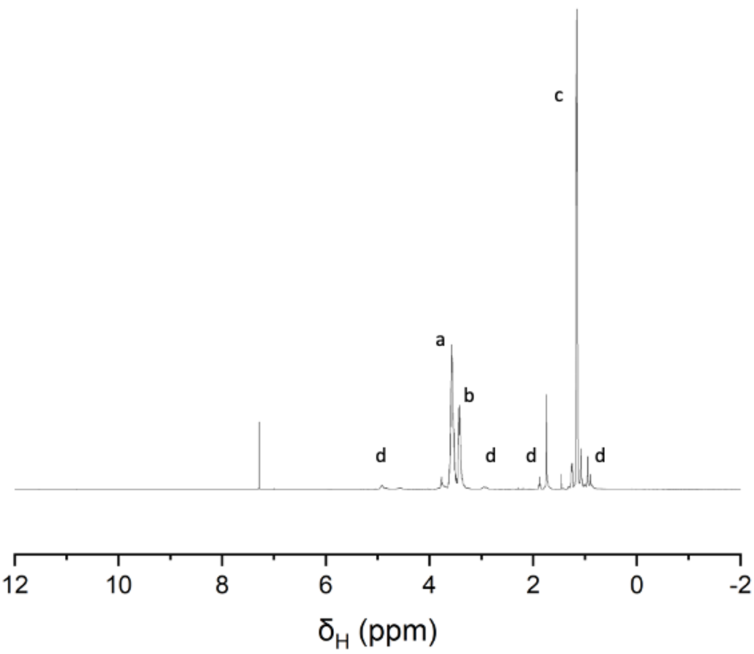


Figure 1 - 1H NMR (400 MHz) of Step Growth Polyurethane

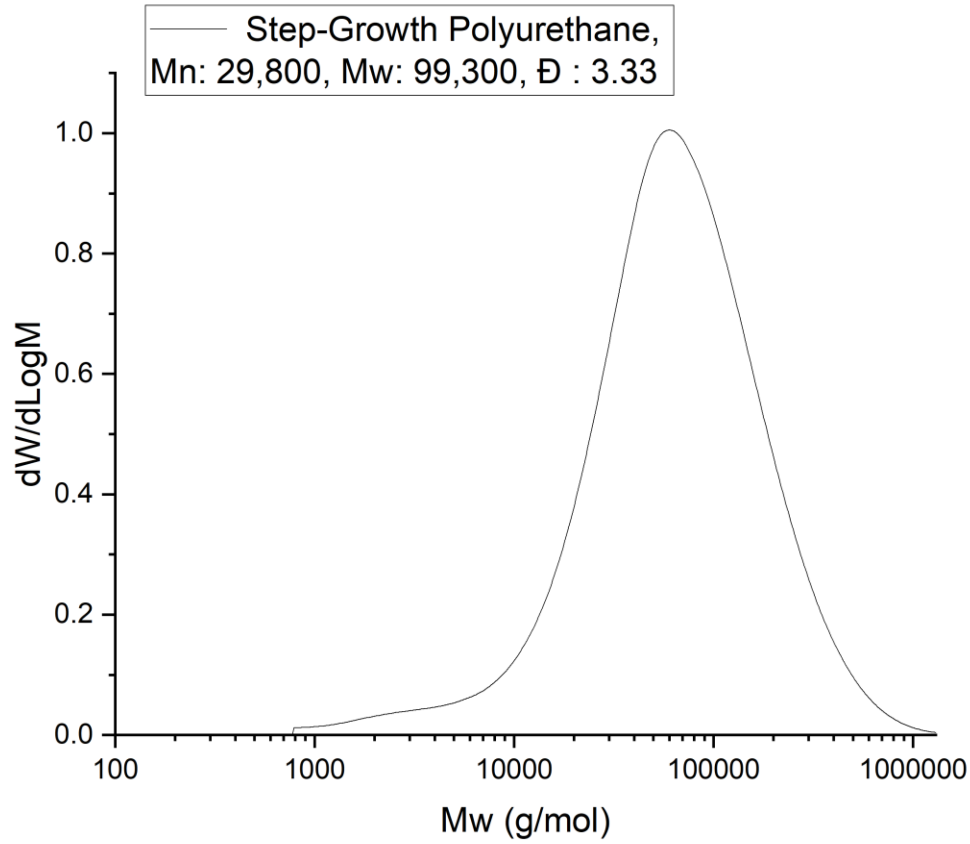
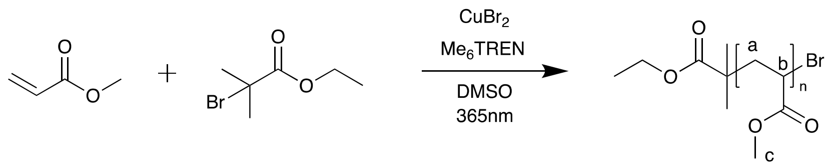


Figure 2 - GPC of Step Growth Polyurethane

Cu-RDRP Synthesis of Poly(Methyl Acrylate)

Methyl acrylate (5ml, 5.55 mmol, 25 eq)., EBiB (326 µl, 2.22 mmol, 1 eq.), $CuBr_2$ (9.90 mg, 44.4 µmol, 0.02 eq.), Me_6TREN (71 µl, 0.27 mmol, 0.12 eq.) and DMSO (5 ml) were added to a round bottom flask that was septum-sealed and degassed with nitrogen for 15 minutes. DMSO (in excess) was added to a separate flask that was also septum-sealed and degassed with nitrogen for 15 minutes. The DMSO was used to flush tubing on a Vapourtec E-Series containing a 365 nm photo-reactor. The polymerisation started once the reaction mixture was flowed through the reactor. The product was obtained as a colourless semisolid. Conversion (NMR) 90%, SEC (THF): M_n = 2500 Da, M_w= 2900 Da, $Đ_M$ = 1.19, MaDDOSY M_{wt} = 2200. SEC Values obtained using PMMA calibration on RI detector.



Scheme 2 - Cu-RDRP Synthesis of Poly(Methyl Acrylate)

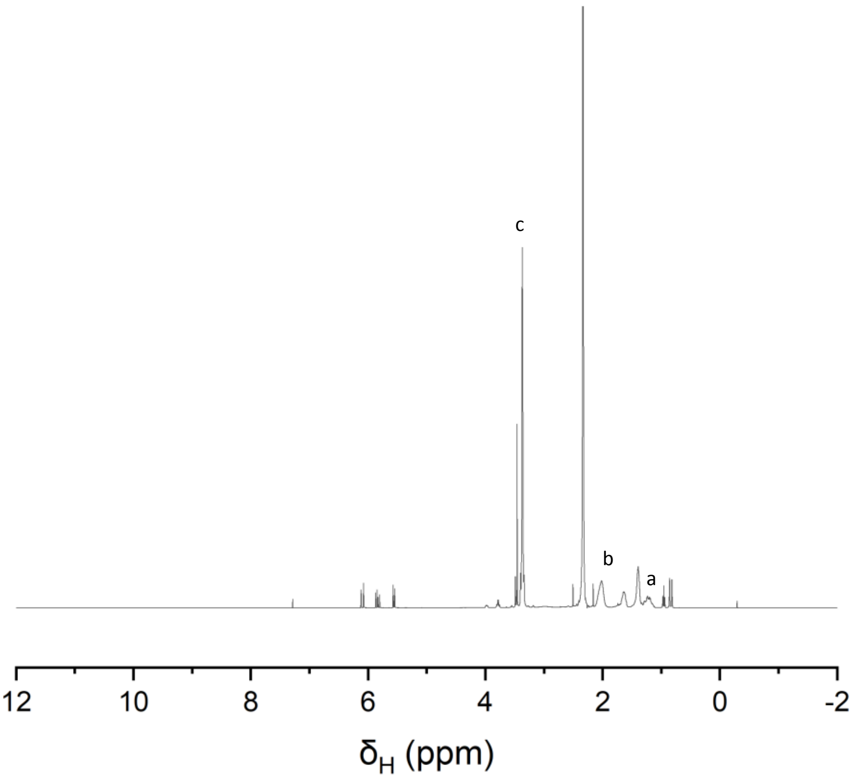


Figure 3 - 1H NMR (400 MHz) of Cu-RDRP of Poly(Methyl Acrylate)

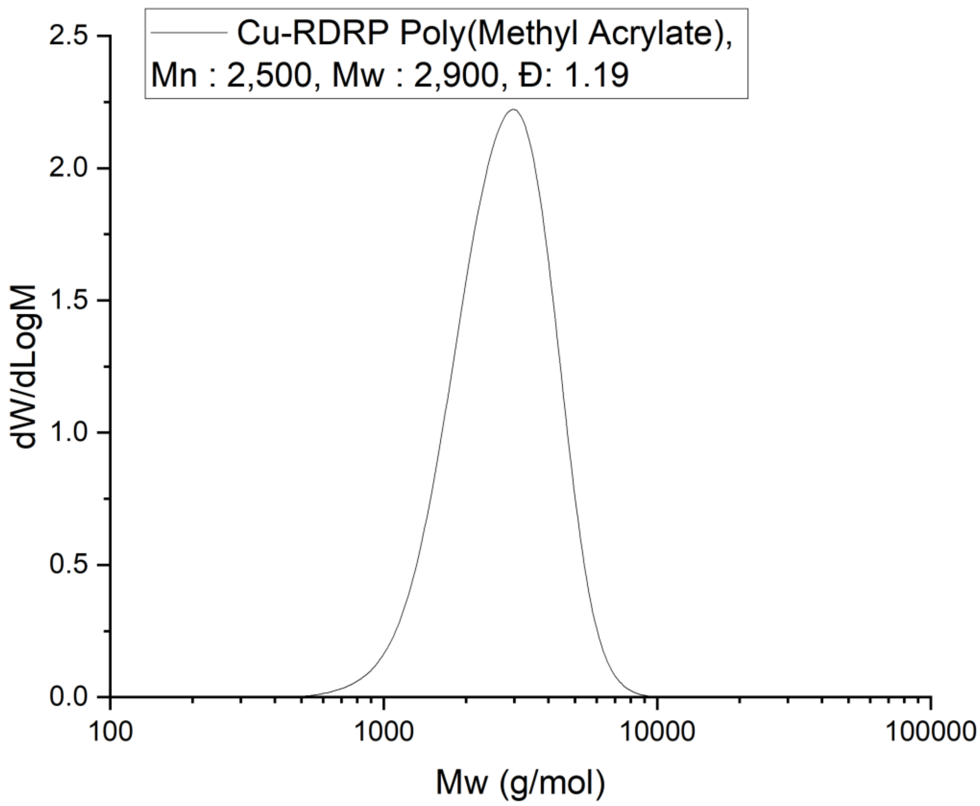
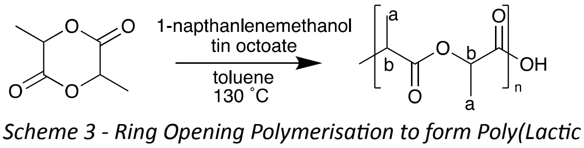


Figure 4 - GPC of Cu-RDRP of Poly(Methyl Acrylate)

Poly(Lactic Acid) Synthesis

In a Schlenk tube equipped with a magnetic follower under an inert atmosphere, the initiator 1-naphthalenemethanol (1 eq.) was added and placed under vacuum for 45 minutes. Simultaneously in a separate flask, 10.0 g of L-lactide (12.5, 25 or 50 eq.) was placed under vacuum for 45 minutes to remove any trace volatiles. Subsequently, the L-lactide was transferred to the Schlenk tube containing 1-naphthalenemethanol followed by 4 mL stock solution of tin octoate (0.001 eq. to L-lactide) in anhydrous toluene. The reaction tube was placed in a pre-heated oil bath and stirred for 1 h at 130 °C. After 1 h, the Schlenk tube was removed from the oil bath and allowed to cool down to ambient temperature. The yellowish solid was dissolved in chloroform and precipitated twice in methanol. The volatiles were removed under reduced pressure to yield the products as yellowish solids. SEC was performed in chloroform and NMR in d-chloroform. The following Mark-Houwink constants were used for the analysis of PLA in THF: K = 22.1 dl/g·10^5, α = 0.77



Scheme 3 - Ring Opening Polymerisation to form Poly(Lactic Acid)

Table 2 - Analysis results for ROP Poly(lactic Acid) samples

Sample	DP	Conversion	M_n [kDa]	M_w [kDa]	$Đ_M$	MaDDOSY M_{wt} [kDa]
PLA 2.7k	12.5	98%	2.09	2.70	1.29	2400
PLA 5k	25.0	97%	3.91	5.00	1.28	4600
PLA 9k	50.0	97%	6.40	8.96	1.40	10700

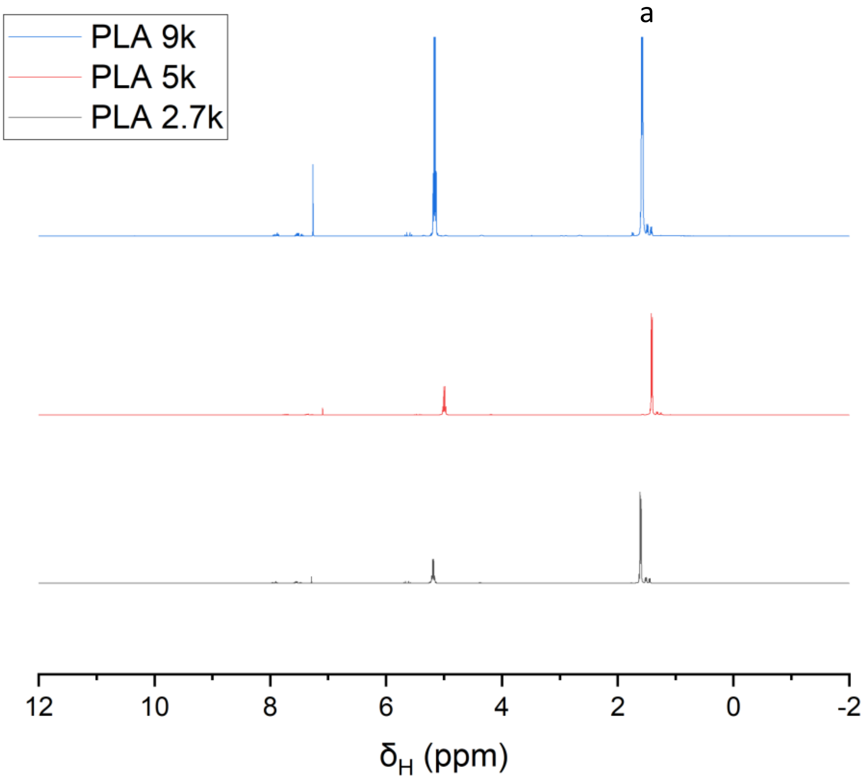


Figure 5 - 1H NMR (400 MHz) of Ring Opening Polymerisation of Poly(Lactic Acid) samples

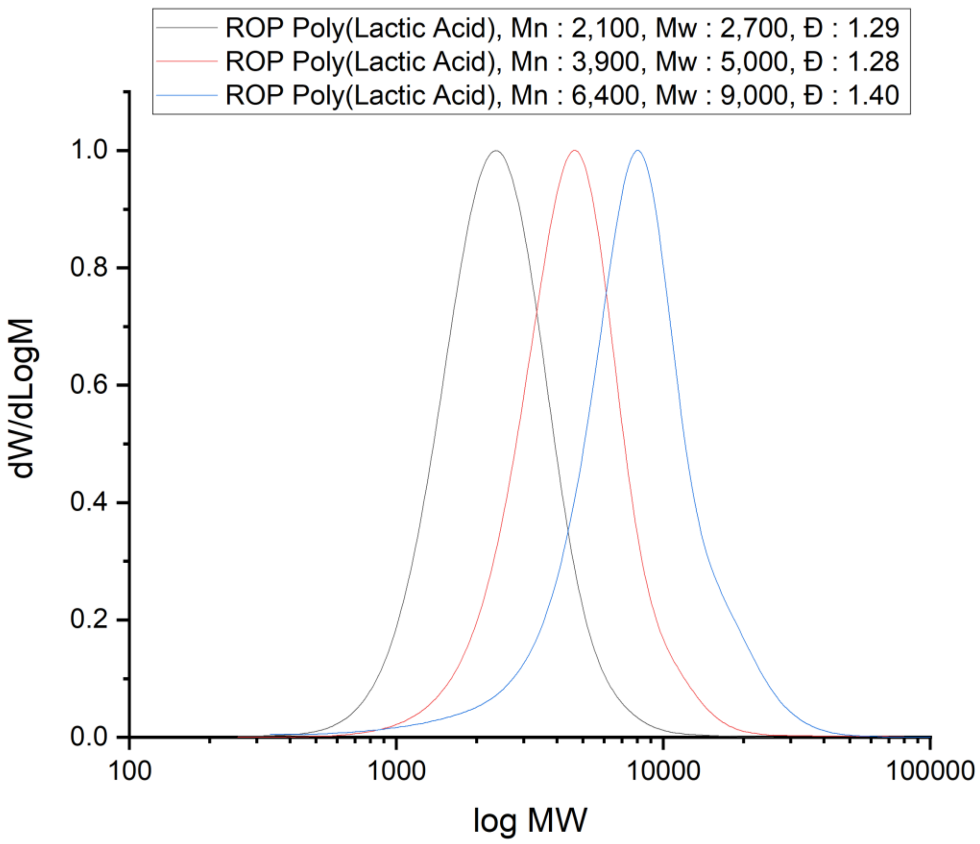
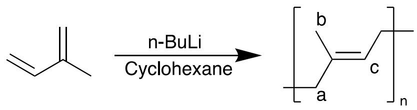


Figure 6 - GPC of Ring Opening Polymerisation of Poly(Lactic Acid) samples

Polyisoprene Synthesis

Isoprene (20.4 g, 0.300 mol, 100 eq.) was dissolved in cyclohexane (270 mL), both of which had been dried over molecular sieves (3 Å), in a flask under nitrogen to give water levels, 8.3 ppm for cyclohexane and 10.7 ppm for isoprene as measured by Karl Fischer Titration. The flask was placed in a reflux setup under nitrogen and left to equilibrate for 20 minutes at room temperature. n-butyl lithium solution (2.10 mL, 1.4 M, 1 eq.) was injected through a rubber septum. The reaction was allowed to proceed at room temperature for 720 minutes. After the reaction, the product was reduced to a colourless, viscous oil with heating. Conversion (NMR) 96%, SEC (THF): M_n = 10,700 Da, M_w = 15,400 Da, $Đ_M$ = 1.15, MaDDOSY M_{wt} = 16,400. SEC Values obtained using PMMA calibration on RI detector.



Scheme 4 - Anionic polymerisation of Isoprene

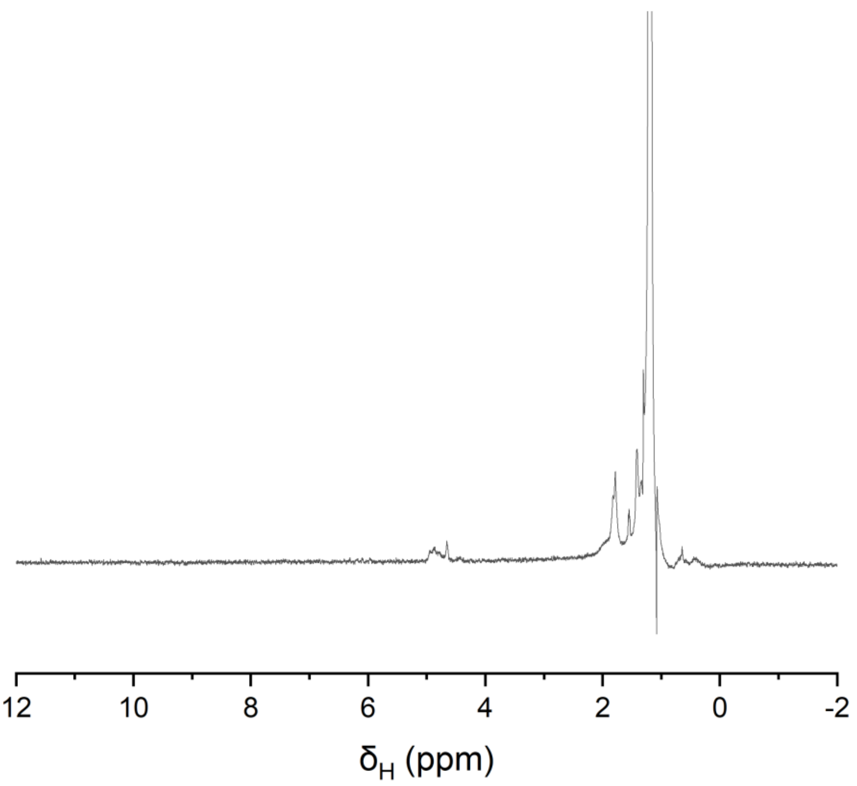


Figure 7 - 1H NMR (80 MHz) of Anionic of Poly(Isoprene)

b

a

c

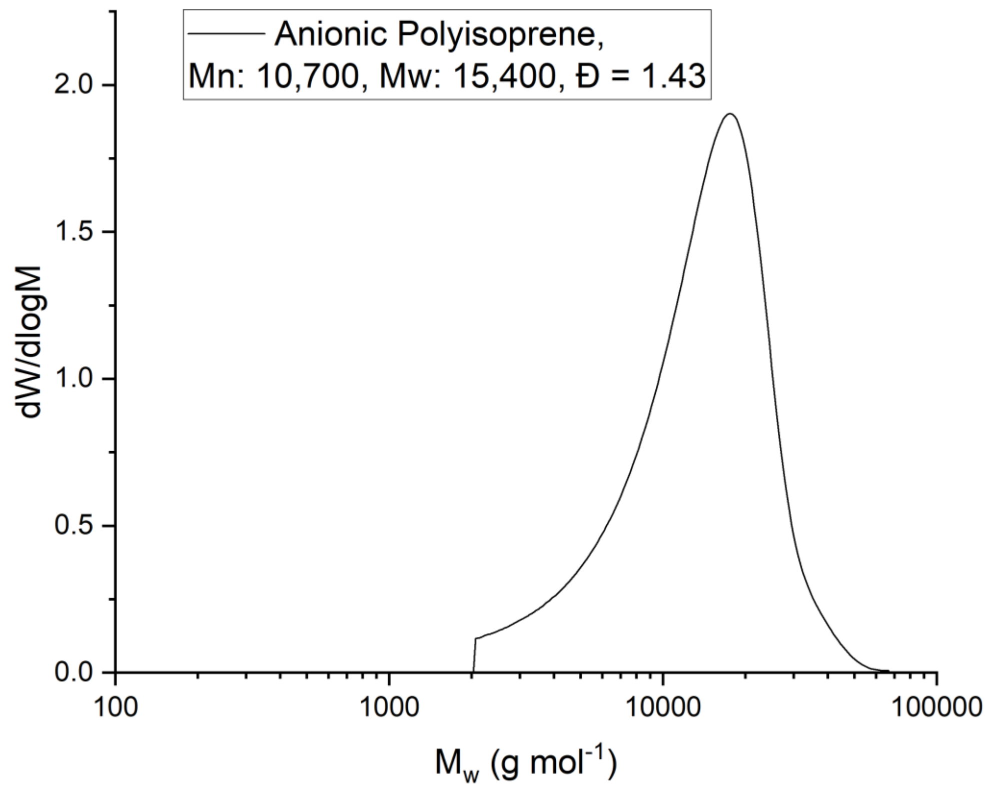
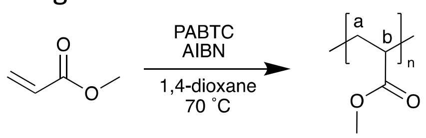


Figure 8 - GPC of Anionic of Poly(Isoprene)

Thermally Initiated RAFT synthesis of Poly(Methyl Acrylate)

Methyl Acrylate (28.5 g, 0.330 mol, 100 eq.) and RAFT agent (789 mg, 3.30 mmol, 1 eq.) were dissolved in 1,4-dioxane (270 mL). The resultant solution was degassed, brought under nitrogen, and heated to 70 °C in an oil bath. AIBN (54.2 mg, 0.330 mmol, 10 eq.) was added and the reaction was allowed to proceed for 810 minutes at 70 °C. After the reaction, the product was reduced to a yellow, viscous oil using a heat gun. Conversion (NMR) 90%, SEC (THF): M_n = 5770 Da, M_w = 7500 Da, $Đ_M$ = 1.30, MaDDOSY M_{wt} = 8100 SEC Values obtained using PMMA calibration on RI detector.



Scheme 5 - Thermally Initiated RAFT Polymerisation of Methyl Acrylate

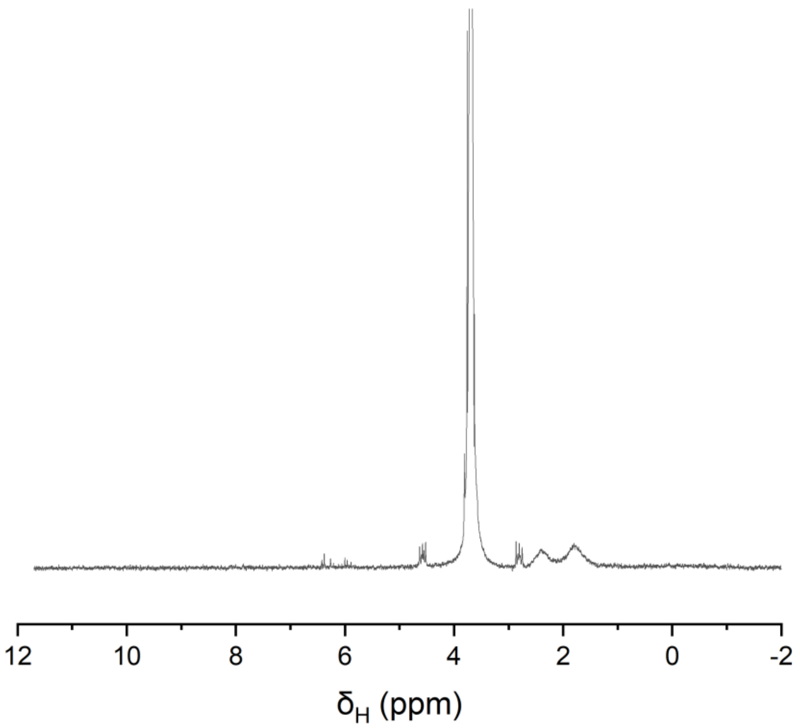


Figure 9 - 1H NMR (80 MHz) of Thermally Initiated RAFT of Poly(Methyl Acrylate)

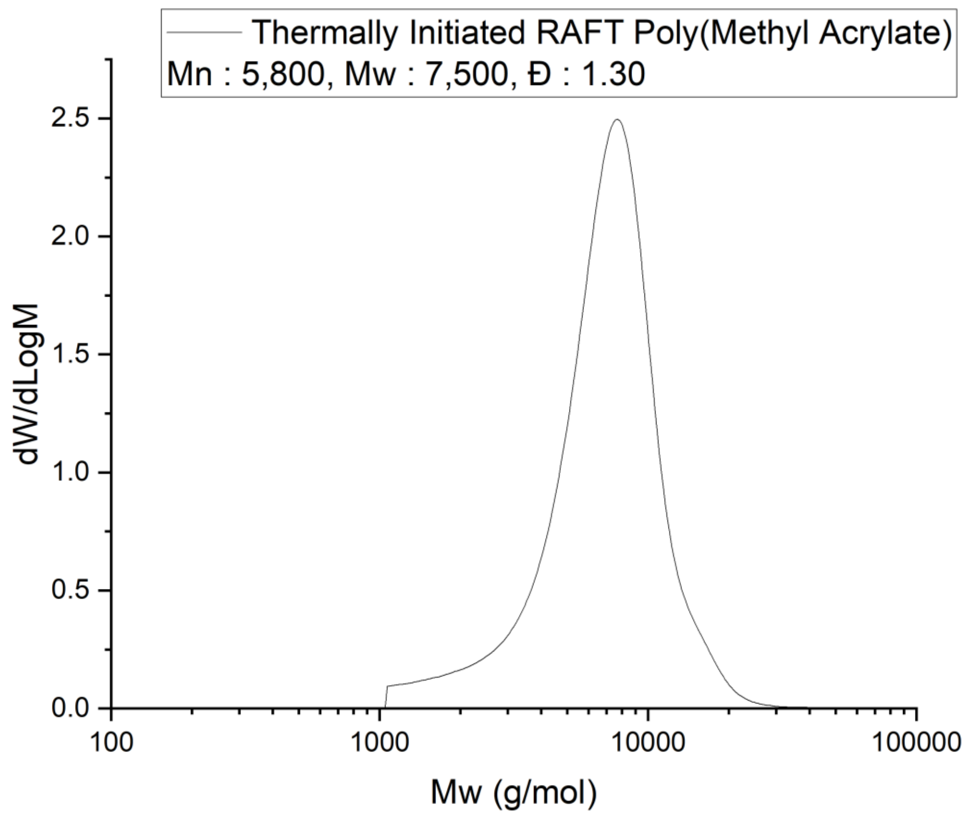
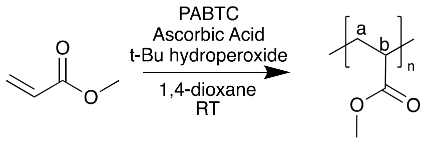


Figure 10 - GPC of Thermally Initiated RAFT of Poly(Methyl Acrylate)

Redox Initiated RAFT synthesis of Poly(Methyl Acrylate)

Methyl Acrylate (28.5 g, 0.330 mol, 50 eq.) and RAFT agent (1.58 g ,6.60 mmol, 1 eq.) were dissolved in 1,4-dioxane (270 mL). The resultant solution was degassed, brought under nitrogen and left to equilibrate for 20 minutes at room temperature. tert-Butyl hydroperoxide (149 mg, 1.65 mmol, 0.25 eq.) and ascorbic acid (145.3 g, 0.825 mmol, 0.125 eq.) were added. The reaction was allowed to proceed for 720 minutes at room temperature. After the reaction, the product was then reduced to a yellow, viscous oil using a heat gun. Conversion (NMR) 68%, SEC (THF): M_n = 3400 Da, M_w = 3800 Da, $Đ_M$ = 1.11, MaDDOSY M_{wt} = 4100. SEC Values obtained using PMMA calibration on RI detector.



Scheme 6 - Redox Initiated Polymerisation of Methyl Acrylate

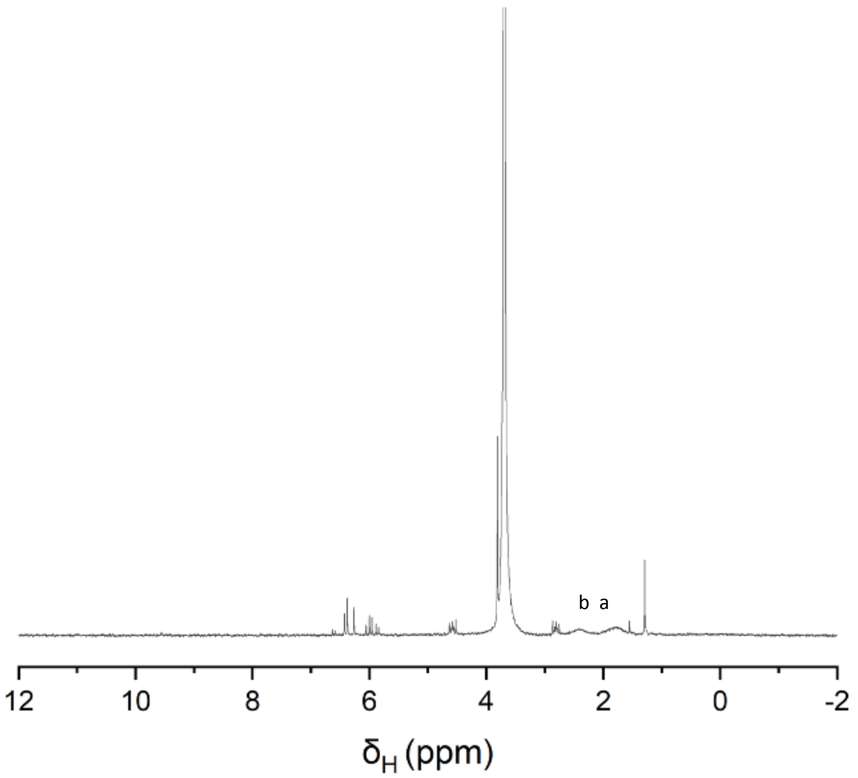


Figure 11 - 1H NMR (80 MHz) of Redox Initiated RAFT of Poly(methyl Acrylate)

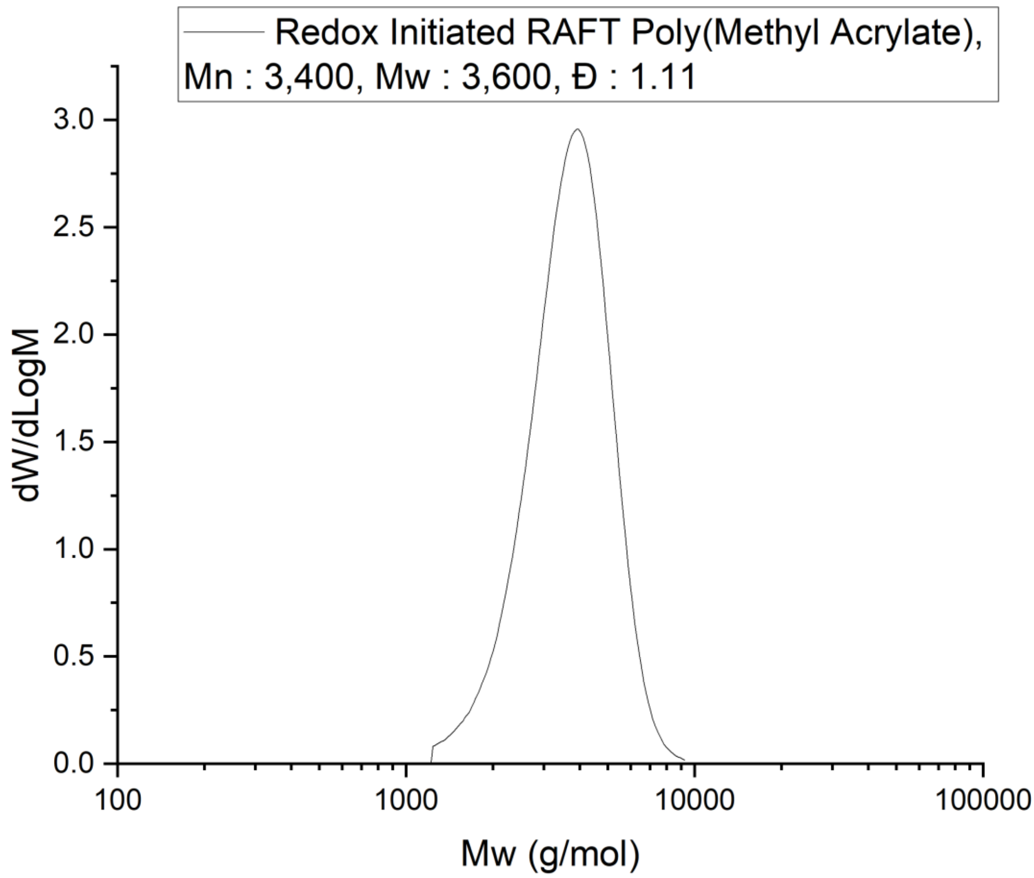


Figure 12 - GPC of Redox Initiated RAFT of Poly(Methyl Acrylate)

Commercial Polymers

Commercial Samples of lysozyme, poly(styrene-co-maleic anhydride) cumene terminated (PSCMA), calcitonin, poly(vinyl pyridine) and styrene-ethylene-butadiene-styrene (SEBS) were each dissolved in the solvents and at the concentrations detailed in table 3, the molecular weights calculated by MaDDOSY and the reported weight average molecular weights(M_w) are also reported in table 3. Weight average molecular weights are as given by the supplier from the certificates of analysis in all cases except the samples of SEBS in which in house GPC was used with Values obtained using PMMA calibration on RI detector.

Table 3 - Experimental details of commercial polymers.

Sample	Solvent	Concentration [mg mL^{-1}]	MaDDOSY M_{wt} [Da]	Reported M_w [Da]
Calcitonin	D_2O	15	4300	3500
Lysozyme	D_2O	15	7700	14300
PSCMA	Acetone-d6	15	3700	3300
Poly(vinyl pyridine)	Dimethylformamide	7	50900	63700
SEBS Polymer Source	$CDCl_3$	10	83900	85600

SEBS Kraton	$CDCl_3$	10	79000	82700
SEBS Sigma-Aldrich	$CDCl_3$	10	84200	84100

Representative Spectra

When processing DOSY data through Magritek Spinsolve version 2.1.3, diffusion coefficients are calculated through a Stejskal-Tanner plot. An example of the stacked spectrum is shown in figure 13, with the resultant Stejskal-Tanner plot shown in figure 14. In both cases here, the figures are for a sample of 100 kDa polyisoprene in $CDCl_3$. In this experiment, the acquisition parameters were as follows: Number of scans = 8, acquisition time = 6.5 s, repetition time = 10 s, little delta = 10 ms, big delta = 100 ms, maximum gradient = 392.25 mT (=75% maximum gradient), number of steps = 8, total experiment time = 10 minutes 41 seconds. The Stejskal-Tanner plot gives a diffusion constant of 2.323×10^{-11} m^2s^{-1}

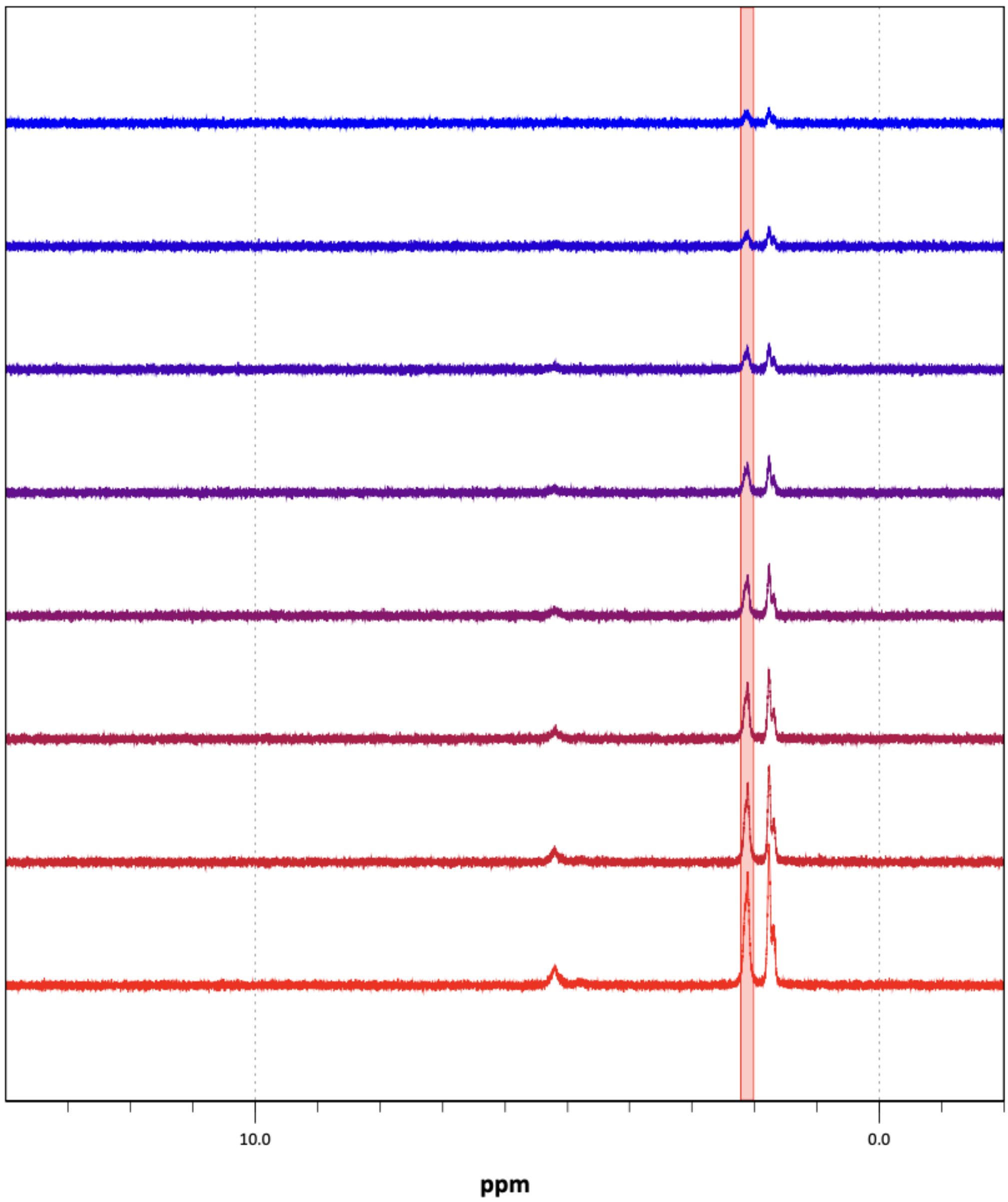


Figure 13 - Stacked PGSTE plot for 100 kDa polyisoprene in $CDCl_3$

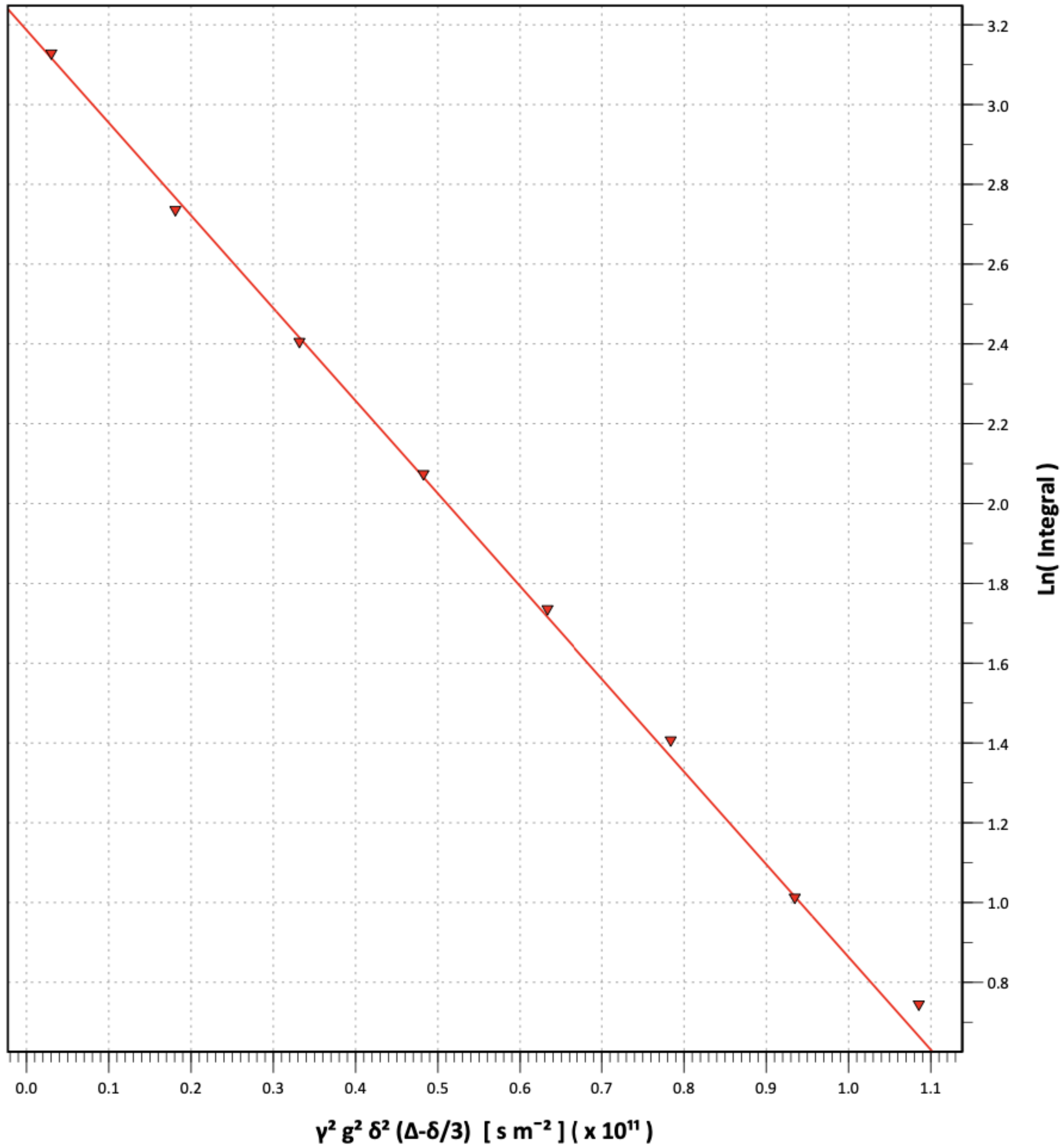
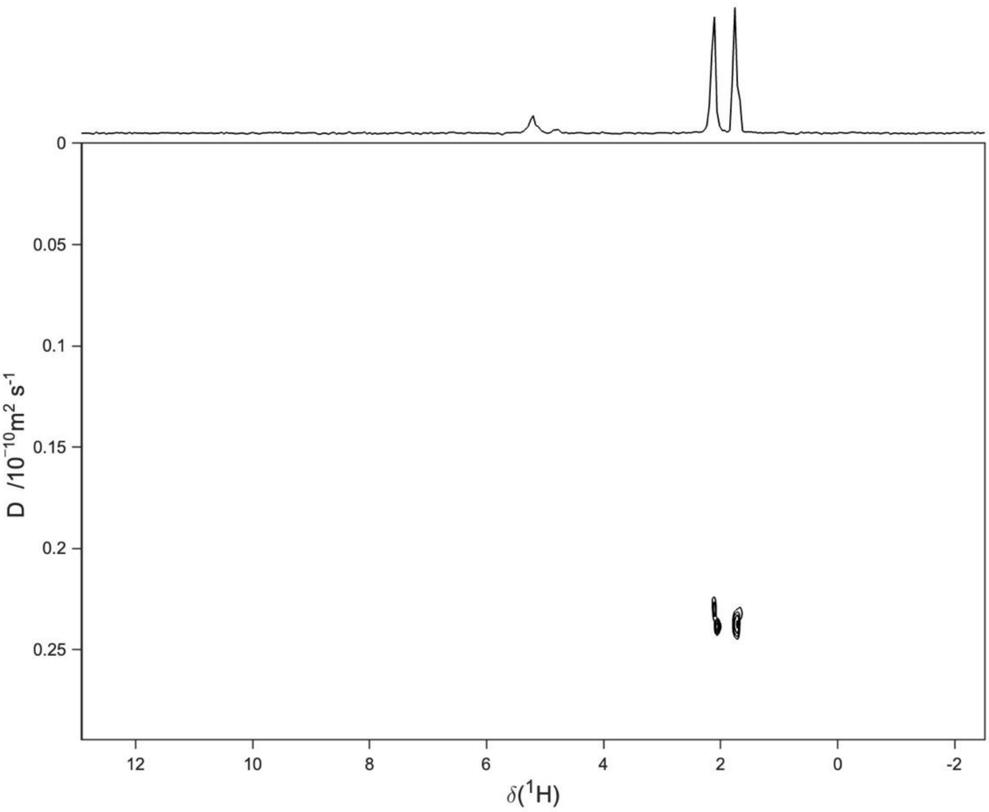


Figure 14 - Stejskal-Tanner plot of 100 kDa polyisoprene in $CDCl_3$

Alternatively, DOSY spectra were also processed using the GNAT, version 1.3.2. In this case a Stejskal-Tanner is again used, and the result is a 2D DOSY spectrum, rather than a stacked spectrum and fitted curve. An example plot of the same 100 kDa polyisoprene dataset as discussed above is shown in figure 15. In this case, the diffusion constant is read as 2.365×10^{-11} $m^2 s^{-1}$. The values from both approaches vary by only 1.8% and therefore it can be assumed either approach can be used.



These spectra are representative of the spectra used throughout this study, individual datasets can be accessed on the Warwick research archive portal: https://wrap.warwick.ac.uk/.

Propagation of Uncertainty of Molecular Weight Calculation

Uncertainty within Molecular Weight Measurement

The calibration allows us to estimate the molecular weight of a given polymer within this range as described by equation 1.

$$M_{wt} = 10^{\frac{(\log D + \log \eta) + 7.74}{-0.597}}$$

Equation 1 - Calculation of Molecular Weight from Diffusion Coefficient and Solvent Viscosity as Described by the Calibration Shown in Figure 2.

However, it is unrealistic to assume this is the exact molecular weight. Indeed, with respect to polymers, a distribution of molecular weights is a statistical certainty. Where techniques such as mass spectrometry may be able to separate the component molecular weights to give a distribution, techniques such as DOSY that rely on measuring a parameter average across an entire sample cannot, and so knowing the uncertainty within the molecular weight estimation is a must.

There are a few approaches that can be used here. Estimating uncertainty from a calibration curve can be done with relative ease when using a normal calibration curve of known standards of the same type, this is done using equation 2,

$$\sigma_{\log(M_{wt})} = \frac{\sigma_y}{m}\sqrt{\frac{1}{k}+\frac{1}{n}+\frac{(y-\bar{y})^2}{m^2\sum_{i=1}^{n}(x_i-\bar{x})^2}}$$

Equation 2 – Standard Deviation in DOSY M_{wt}.

Where $\sigma_{\log(Mwt)}$ is the standard deviation in the log of the estimated molecular weight, σ_y is the standard deviation in log (D) + log (η) in the calibrants, m is the slope of the calibration curve, k is the number of scans in the sample DOSY experiment, n is the number of points in the calibration curve, y is the meased log (D) + log (η) of the sample, $\bar{y}$ is the mean of log (D) + log (η) in the calibrants and $\sum_{i=1}^{n}(x_i-\bar{x})^2$ is the sum of the squared deviations from the mean of the calibrants. This equation is adapted from the work by Hibbert on 'The uncertainty of a result from a linear calibration'.[4]

The standard error about the mean for a measurement can be related to the standard deviation calculated above through equation 3:

$$SE_{\log(M_{wt})} = \frac{\sigma_{\log(M_{wt})}}{\sqrt{k}}$$

Equation 3 - Standard Error in DOSY M_{wt}.

This standard error can be related to 95% confidence intervals, Int, in $\log(M_{wt})$ through equation 4[5]. It is worth noting here that we assume the value in question is normally distributed and as such this equation holds true, in reality this may not be the case, however this approach remains a good approximation of the confidence intervals.

$$Int \; = \log(M_{wt}) \pm (1.96 \times SE_{\log(M_{wt})})$$

Equation 4 - 95% Confidence Intervals from Standard Error.

A worked example of calculation of the confidence intervals for the anionic polyisoprene sample is shown in scheme 7.

$$(1) D = 2.47 \times 10^{-11}\ m^2 s^{-1}$$

$$(2) \log_{10} D + \log_{10}\eta = -10.257$$

$$(3)\ \sigma_{\log(M_{wt})} = \frac{0.520}{-0.597}\sqrt{\frac{1}{8}+\frac{1}{31}+\frac{0.0271}{0.356 \times 21.930}} = 0.349$$

$$(4)\ SE_{\log(M_{wt})} = \frac{0.349}{\sqrt{31}} = 0.0626$$

$$(5) \log_{10}(Int) \; = 4.209 \pm (1.96 \times 0.0626)$$

$$(6)\ M_w = 12200 < \mathbf{16400} < 22100\ g\ mol^{-1}$$

Scheme 7 - Calculation of confidence intervals in estimated molecular weight

Another approach is to first calculate a standard error for the polymer of interest against each calibration polymer type, using equations 2 and 3, and then weight the results using equation 5. The confidence intervals can then be found using equation 4. The results obtained from both methods are shown in table 5.

$$w\sigma_{\log(M_{wt})} = \frac{\sum_{i=1}^{n} n_i \sigma_{\log(M_{wt})_i}}{\sum_{i=1}^{n} n_i}$$

Equation 5 - Weighting of RMSE Values

Where n_i is the number of polymer type i and $\sigma_{\log(M_{wt})_i}$ is the root mean squared error of polymer type i. Confidence intervals can then be found as ±2 times the RMSE[6] Likewise, molecular weights can be calculated using calibration curves of each polymer type, and weighting the results. As an example, for the anionic polyisoprene sample, the molecular weights and standard deviations calculated against each polymer type in the calibration are given, along with the weighted molecular weights and standard deviations, calculated using equation 5 in table 4:

Table 4 - Parameters used during method 2 of calculation for anionic polyisoprene sample

Polymer Type	Calculated molecular weight / g mol^{-1}	$\sigma_{\log(M_{wt})}$
PAMAM	11813	0.21
Polyisoprene	13587	0.14
PMMA	13388	0.36
Polystyrene	11633	0.27
Poly(styrene sulfonate)	31901	0.24
PEG	12310	0.29
Weighted Averages	14651	0.24

The standard deviation can then be propagated into upper and lower bounds, as detailed above, and shown in table 5.

Table 5 - Calculations of Molecular Weight and Confidence Intervals by MaDDOSY

Polymer	Calculation using Equations 2, 3 and 4	Calculation using Equations 2-6	GPC M_w
Polyurethane	72300<**101000**<142000	17200<**105200**<645000	99300
Cu-RDRP MA	1540<**2160**<3030	680<**2040**<6130	2910
ROP PLA 2.7k	1720<**2400**<3360	780<**2300**<6750	2700
ROP PLA 5k	3370<**4630**<6350	1620<**4450**<12200	4990
ROP PLA 9k	8050<**10800**<14400	3800<**10700**<30000	8960
Anionic Polyisoprene	12200<**16400**<22100	5040<**14700**<42600	16500
Thermal RAFT MA	6050<**8120**<10900	2830<**7800**<21500	7510
Redox RAFT MA	3000<**4060**<5470	1380<**3830**<10600	3810
Calcitonin	3140<**4320**<5940	1510<**4150**<11440	3460
Lysozyme	6440<**7680**<9150	3120<**7480**<17900	14300
PSCMA	2690<**3720**<5130	1280<**3560**<9910	3270
Poly(vinyl pyridine)	40200<**50900**<64600	14000<**51800**<191700	63700

SEBS Polymer Source	67500<**83900**<104300	20200<**86300**<367700	85600
SEBS Kraton	63400<**79000**<98400	19400<**81100**<339300	82700
SEBS Sigma-Aldrich	67800<**84200**<104700	20300<**86600**<369600	84100

In each case, the number in bold is the predicted molecular weight, and the numbers flanking it are the lower and upper 95% confidence intervals, respectively. Universally, we see excellent agreement between the predicted molecular weight and the GPC weight average molecular weight. We have elected to use the first method when calculating uncertainties moving forward, allowing us to calculate molecular weights and uncertainties across the entire range of calibrants, so as not to put more weighting on those polymer types which had more calibration standards available.

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